

Borel-Torsor Rosser Phantom

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Summary

Pass 89 states the Rosser phantom as a single Borel-torsor / extension-class theorem. In the APS/Rosser model developed in the repository, the Guaspari-Solovay witness-comparison Čech class, the $\varprojlim^1(\mathbb{Z}, \times m)$ phantom, the all-prime quotient $\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$, the finite-adele extension

$$0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0,$$

and the hyperbolic Borel shear orbit for $\mathbb{Q}^\times \ltimes \epsilon$ are presentations of the same torsor class.

Main Claims

- **Witness-to-phantom bridge.** A Rosser witness-comparison Čech cocycle maps to its class in $\text{coker } \delta$, identified with $\widehat{\mathbb{Z}}_m/\mathbb{Z}$ or ϵ .
- **Phantom-to-adele bridge.** Pushing out $0 \rightarrow \mathbb{Z} \rightarrow \widehat{\mathbb{Z}} \rightarrow \epsilon \rightarrow 0$ along $\mathbb{Z} \rightarrow \mathbb{Q}$ gives the finite-adele extension $0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0$.
- **Borel level.** Strict integral marking is rigid; forgetting it leaves the Levi $\mathbb{Q}^\times$; the full $\mathbb{Q}^\times \ltimes \epsilon$ appears only in the hyperbolic realization $H = \epsilon \oplus \mathbb{Q}$.
- **Gauge invariance.** Changing Guaspari-Solovay witness choices changes representatives, sections, and finite Loeb lifts by coboundaries, but it preserves the torsor class, finite conductor restrictions, radical support, and finite-adele extension line.

Machine Verification

`code/scripts/check-pass89.py` produced `artifacts/reports/pass89-borel-torsor-rosser-phantom-check.json` with overall PASS. It checks finite Čech windows where representatives change but classes are preserved, affine Borel shadows $(\mathbb{Z}/N)^\times \ltimes \mathbb{Z}/N$, singleton strict marked stabilizers, and the invariant/non-invariant split under witness changes.

Next Use

Pass 90 should make this torsor theorem functorial across conductor and radical maps, comparing m -adic variants with the all-prime ϵ case in one natural diagram.